



Mohd Affan Khan
Roll No.: 244103111
M.Tech - Mechanical Engineering
Specializing in Computational Mechanics
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Mechanical Engineer with expertise in Computational Mechanics including CFD, FEM and Optimization, combining ML-driven design and a granted 3D printing patent to deliver innovative engineering solutions.

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech	Indian Institute of Technology, Guwahati	8.82	2024-2026
B.Tech	G.B. Pant Govt. Engineering College	7.3	2017-2021
Senior Secondary	CBSE Board	86.4 %	2017
Secondary	CBSE Board	9.0	2015

EXPERIENCE

- R&D Engineer** Jul. 2026 – Present
SEDEMAC Mechatronics Pune
– Designing of Electric Machines.
- Project Scientist** Oct. 2023 – Jul. 2024
Indian Institute of Technology, Delhi Delhi
– Designed and validated a Coriolis mass flow meter with 99.9% measurement accuracy for high-precision industrial flow-sensing applications.
- Project Associate** Jul. 2022 – Jun. 2023
Indian Institute of Science, Bangalore Bengaluru
– Fabricated a pneumatically actuated resin 3D printer with a flexible printing arm, enabling in-space manufacturing applications.
- Design Engineer** Aug. 2021 – Jun. 2022
Tow-Botics Systems WFH
– Developed an underwater ship maneuvering robot and performed FEA in ANSYS for structural iterative optimization and static validation, achieving 2× increase in load-carrying capacity.

PROJECTS

- Development of Generative Structural and Fluid-Thermal Topology optimization model** Jul. 2025 – present
Prof. D. Sharma / M.Tech Project [View](#)
– Built a generative design framework using a 2D Darcy-based thermal–fluid solver and deep-learning surrogates, achieving an R² score of 0.9998 on structural topology-optimized generated datasets.
- Active Flexible Compliant-Based Bionic Arm** Jan. 2025 – Apr. 2025
Prof. S. Kanagraj / Biomedical Devices Course Project [View](#)
– Developed a lightweight, compliant-based bionic arm using 3D printing, ESP32 control, and muscle-sensor input. Designed the mechanism through CAD and nonlinear ANSYS analysis, implemented PD control for fast, accurate motion, and built a functional low-cost prototype.
- SAE ISS Aero Design Challenge** Oct. 2019 – Mar. 2020
B.Tech Project [View](#)
– Engineered a 2m wingspan aircraft prototype capable of carrying a 5 kg payload, designing the complete fuselage and wing assembly in SOLIDWORKS and iterative optimization of aerodynamic performance using CFD analysis.

TECHNICAL SKILLS

- Programming:** Python (NumPy, Pandas, Scikit-learn, Tensorflow, Keras), C, MATLAB.
 - Analytical & Computational:** Machine Learning (Regression, CNN, GANs), Optimization (Multi-variable, Constrained), Numerical Methods.
 - CAE Software:** ANSYS Workbench: (Fluent, Static Structural, Modal, Harmonic, Composite PrepPost (ACP)*).
 - CAD Tools:** SolidWorks, PTC Creo, ANSYS SpaceClaim, Catia V5*.
- * Elementary proficiency*

KEY COURSES TAKEN

- Finite Element Method
- Optimization methods in Engineering
- Numerical Analysis
- Robotics and Robot Applications
- Computational Fluid Dynamics
- Rotor Dynamics
- Advanced Solid Mechanics
- Continuum Mechanics
- Computational Mechanics Lab

POSITIONS OF RESPONSIBILITY

- Department Placement Representative,** M.Tech Computational Mechanics, Dept. of ME May. 2025 - Present

ACHIEVEMENTS, CERTIFICATIONS AND PATENT

● Indian Patent (No. 589444) <i>Inventor on a granted patent for a Polymer 3D Printer with a Cable-Actuated Flexible Arm, enhancing precision and flexibility.</i>	<i>Jul. 2025</i>  View
● Siemens (Coursera) <i>Certified in Applied Computational Fluid Dynamics (CFD), focusing on executing practical simulations and industry applications. (95.33%)</i>	<i>Sep. 2025</i>
● Dassault Systèmes <i>Attained Certified SOLIDWORKS Associate (CSWA) credential, demonstrating proficiency in 3D CAD modeling. (240/240 Score)</i>	<i>Oct. 2022</i>
● SAE ISS Aero Design Competition <i>Secured All-India Rank 27 for innovative aircraft design and effective project execution.</i>	<i>Mar. 2020</i>